

New PID Parameters Tuning Formulae for Relay Feedback Self-tuning

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Abstract—This paper presents the way to derive tuning formulae for PID parameters used to relay feedback self-tuning. The method is based on the parameter optimization technology of minimizing the IAE criteria and the technology of data fitting. A simulation example has demonstrated the validity of the new tuning formulae, and the comparison between the new tuning formulae and the ZN tuning formulae and the tuning formulae based on gain margin and phase margin illustrates the priority of the new tuning formulae.

Keyword: relay feedback, tuning formula, IAE, curve fitting;

I. INTRODUCTION

Astrom and Haggglund proposed the original relay feedback self-tuning method in 1984, namely adding a relay module to the close-loop systems to obtain critical information. The critical parameters K_u and T_u are derived by oscillation curve, and the PID parameters are calculated by Z-N tuning formulae. The control object is unknown in the whole process, the identification and tuning are simple, and the self-tuning method based on relay feedback also works well for the highly nonlinear objects.

The self-tuning method based on relay feedback uses the ZN tuning (TABLE 1) formulae to derive the PID parameters. The basis of ZN tuning formulae is that the step response reaches quarter damping ratio. However, the only pursuit of control performance criteria does not actually fit for some conditions, for example on the condition of where we need little overshoot^[1]. Therefore, several tuning formulae came up.

In 1991, Hang and Astrom proposed the conception of normalized delay time constant and first-order time constant,

and modified the controller's structure, and gave the modified ZN formulae (RZN for short).the Cohen-Coon method based on FOPDT(first-order plus delay time) plant reaches quarter damping ratio by dominant pole assignment to obtain the optimum parameters. The KLV method is applied to the plant that has a small time-delay ratio. This method needs the frequency information of point ω_{180} or ω_{135} . There are also some other methods such as ITSE (integral time squared error) optimum tuning method and tuning formulae based on gain margin and phase margin. Many other methods are extended from the methods above. When attempts are made to tune the PID parameters, it is suggested that choose the proper method considering the different plants, properties, prior knowledge and control performance requirement^[2,3]. The paper proposes a new method to derive PID parameters tuning formulae for relay feedback self-tuning, which is mainly based on the optimization technology of minimizing the IAE criteria and the data process technology of curve fitting.

II. THE SELECTION OF INTEGRAL PERFORMANCE CRITERIA

There are three kinds of common performance criteria: ①quality criteria of settling process, as overshoot, settling time, rising time, etc; ②positive definite quadratic form integral function; ③integral performance criteria, namely a function of control system error, including IE, IAE, ITSE, ITAE,etc^[4].

The paper picked the absolute error integral function (IAE) as the performance criteria to derive the tuning formulae, since this criteria emphasizes the overshoot and rapidity of system response. Therefore, the IAE criteria is suitable for the conditions where little overshoot and short settling time are needed, such as the condition of temperature control. IAE can be easily calculated by computer. The process time T should be proper, the system can result in strong robusticity and week sensitivity if T is large, in contrary, the robusticity would be weaker if T is small. In the practical application, the cut and try method is recommended^[5].

TABLE1. THE ORIGINAL ZN TUNING FORMULAE

Controller	K_p	T_i	T_d
P	$K_u/2$		
PI	$K_u/2.2$	$T_u/1.2$	
PID	$K_u/1.7$	$T_u/2$	$T_u/8$

III. THE DERIVE PROCESS OF PID PARAMETERS TUNING FORMULAE USED TO RELAY FEEDBACK SELF-TUNING

Firstly, two different types of control objects which are common used in simulation were chosen: first-order plus delay time (FOPDT): $K \times e^{-\tau s} / (1 + T_s)$ and high order inertia link: $K / (1 + T_s)^n$, secondly, the control object parameters were changed and the critical information $\{K_u(i), T_u(i), i=1,2,\dots\}$ corresponding to the every changed object was obtained by relay feedback test. Thirdly, a searching program was designed. The optimum PID parameters corresponding to the every changed object which can minimize the IAE criteria could be found by the searching program. Fourthly, the critical information and the optimum PID parameters were put in pairs with the form likes $\{K_u(i), T_u(i), K_p(i), T_i(i), T_d(i), i=1,2,\dots\}$. Fifthly, the structure of new formulae was designed artificially, and the tuning formulae parameters were fitted with the dataset which was consist of critical information and optimum PID parameters.

Taking advantage of the simulink's visualization and easy operation, a PID control simulation experiment platform (Fig.1) was put up in simulink. The system contained a module to calculate IAE value.

The PID parameters optimization program was compiled on matlab. This program could call the PID control simulation system such as figure.1 circularly. Three PID parameters K_p, T_i, T_d were replaced by three variables $p(o), i(w), d(q)$. L, M and N were the length of the three variables respective, and a three-dimensional matrix R was defined. Then the three variables $p(o), i(w), d(q)$ in workspace were changed circularly by three nested-loops, and the matrix R was used to save the IAE value which was obtained from the PID control simulation experiment with each group of three parameters $p(o), i(w), d(q)$. The PID parameters calculated by ZN tuning formulae were used to be the initial value to cycle the three variables $p(o), i(w), d(q)$, the optimum PID parameters were searched by exhaustive method in condition of that there was no prior knowledge. The parameter search had two steps. In the former step, the step size of integral value was ten while proportion and differential's were one on the basis of that the integral value is almost several times of the proportion value and differential value. In the later step, the optimum parameters of the former step were used to be the initial value, of which more accurate optimum parameters were obtained on the foundation [6]. After finishing the program circulation, all

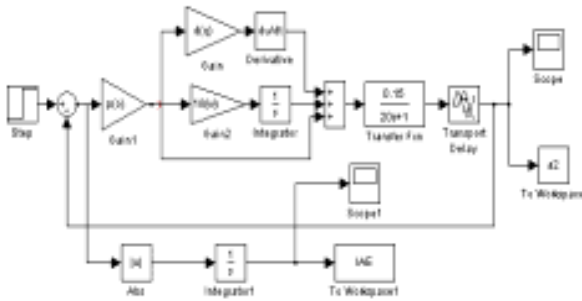


Figure1. PID control simulation experiment platform

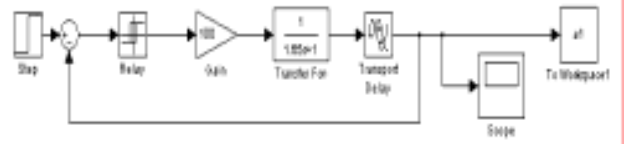


Figure2. Relay feedback experiment systems

the IAE values were saved in the matrix R . The minimal IAE value and its corresponding coordinate were found by another matlab program which was used to search the minimal value in matrix. Later, the optimum PID parameters were obtained [7].

A relay feedback experiment (Fig.2) was put up in simulink. When performing the experiment, the relay feedback experiment oscillation wave with the form of array was saved to matlab's workspace. The wave peak amplitude a_0 and the oscillation frequency ω_0 were measured. And then critical information K_u and T_u were calculated by $K_u = 4h / \pi a_0$, $T_u = 2 \pi / \omega_0$.

The critical information and the optimum PID parameters were put in pairs to be the data source to tuning formulae parameters fitting. The data were derived from 21 functions, 15 functions were FOPDT model and 6 functions were high order inertia link model. The data were fitted by fitting software called 1stopt to obtain the new tuning formulae (table.2). The software supports almost 1000 model structure. The user can build new model too and the software was only used to fit parameters. The model structure in table 2 was chosen artificially according to the principles of simple structure and highly fitting. There separated fitting degrees were 95.7%, 97.7% and 98.9%.

IV. EXAMPLE VERIFICATION

$G(s) = e^{-0.99s} / 1.65s + 1$ was supposed to be the control model [8]. The critical information $K_u=2.87$ and $T_u=3.23$ were calculated by the system in Fig.2 and $K_p=1.14$, $T_i=2.17$, $T_d=0.27$ were calculated by the new tuning formulae in TABLE 2. The step response with the above PID parameters was indicated by real line in Fig 3. The same model was used to be simulated for the comparison with the parameters which were obtained by the ZN formulae and the method based on gain margin and phase margin, the step response were indicated by dashed line in Fig.3. The Fig.3 showed obviously

TABLE2. NEW TUNING FORMULAE

parameters models	P ₁	P ₂	P ₃
$K_p = p_1 \times K_u^{p_2} \times T_u^{p_3}$	0.430	1.034	0.038
$T_i = p_1 \times T_u e^{p_2 K_u}$	0.668	0.116	
$T_d = p_1 \times T_u e^{p_2 K_u}$	0.083	0.115	

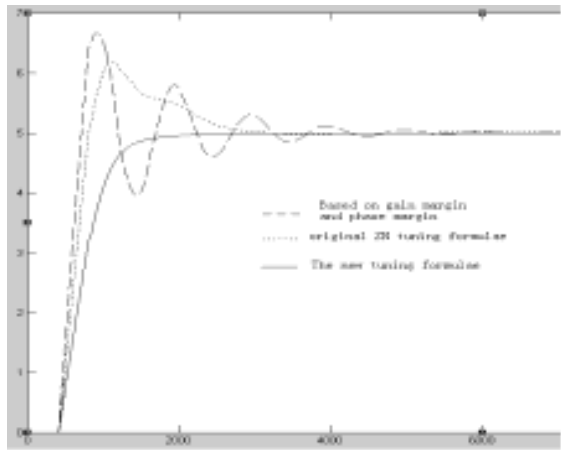


Figure3. The comparison of control system step response

that the step response by the new tuning formulae had no overshoot and short settling time.

V. CONCLUSION

The above example has demonstrated the validity and good performance of the way which was used to derive PID parameters tuning formulae. This way can be extended to apply to more tuning formulae for different models. The

universality of the new formulae has not been proved because of lacking model and data. The problem awaits further study.

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